

# Week 5 — Technical Infrastructure & Modeling

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*"The TCD says what the system must respect. The TMD shows what the system actually looks like at a level an engineer can build from."*

Week 5 takes the **Technical Concept Document (TCD)** from Week 4 and goes one level deeper. Each triad produces a **Technical Modeling Document (TMD)** — a sibling artifact that documents:

- The **data model** the feature reads, writes, and indexes
- The **cloud topology** it deploys to
- The **API contract** it presents to callers
- A **system-level technical model** (sequence diagrams) tying it all together
- A **performance baseline + monitoring plan** that lets the team know whether the SLOs from Week 4 are actually being met
- An **AI-summary validation log** — the discipline of checking AI-generated data summaries against the source

By Friday afternoon, every triad has a TMD that an engineering lead would accept as scoping input for sprint planning.

## What's different about Week 5

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- **The TCD becomes the spine.** Every section of the TMD references a corresponding TCD section.
- **Technical depth increases.** This is the most engineering-adjacent week. TPMs aren't expected to *make* the choices, but they should be able to read engineering proposals and ask sharp questions.
- **AI use is mature.** The Week-2 prompt patterns and Week-4 AI norms apply. New today: explicit **AI-summary validation** discipline (validating AI's read of *real* data, not just text).

## Learning outcomes

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By Friday afternoon, each participant can:

1. Read and reason about a **data model** at the entity level — primary keys, foreign keys, indexes, normalization vs denormalization trade-offs.

2. Frame **cloud topology** decisions (region, availability zones, managed services, multi-tenancy) in customer and cost terms.
3. Author or review a **REST API contract** with sharp resource modeling, idempotency, versioning, and error semantics — and explain when **SOAP** is still the right call.
4. Produce a **sequence diagram** for a feature's happy path that names every hop, every protocol, and every failure handler.
5. Set realistic **performance baselines** anchored to the SLOs from Week 4 and design a **monitoring plan** that surfaces breaches before users do.
6. Validate an AI-generated data summary against source data and produce a **validation log** that documents what was correct, partly correct, and wrong.

## Daily map

| Day | Topic                                                    | Key artifact produced                               |
|-----|----------------------------------------------------------|-----------------------------------------------------|
| 1   | Database structures & data logic                         | TMD Section 1 — Data model                          |
| 2   | Cloud architecture & infrastructure                      | TMD Section 2 — Cloud topology                      |
| 3   | REST & SOAP API fundamentals                             | TMD Section 3 — API contract                        |
| 4   | High-level technical modeling                            | TMD Section 4 — System sequence + integration model |
| 5   | Performance baselines, monitoring, AI-summary validation | TMD Section 5 + final TMD assembly                  |

Each day adds a section. Friday assembles the whole document and runs an integration pass — same discipline as the Week-3 PRD assembly and Week-4 TCD assembly.

## Daily cadence (applies all five days)

| Clock            | Block                                   | Mix                                           |
|------------------|-----------------------------------------|-----------------------------------------------|
| 09:00 –<br>09:15 | Opening & objectives                    | Instructor                                    |
| 09:15 –<br>10:30 | Teaching Block 1 + Activity 1           | ~25 min teach / 50 min triad work             |
| 10:30 –<br>10:45 | <b>Break</b>                            |                                               |
| 10:45 –<br>12:00 | Teaching Block 2 + Activity 2           | ~25 min teach / 50 min triad work             |
| 12:00 –<br>13:00 | <b>Lunch</b>                            |                                               |
| 13:00 –<br>14:15 | Teaching Block 3 + Activity 3           | ~25 min teach / 50 min triad work             |
| 14:15 –<br>14:30 | <b>Break</b>                            |                                               |
| 14:30 –<br>16:00 | Teaching Block 4 + Activity 4 +<br>Wrap | ~25 min teach / 45 min triad / 20 min<br>wrap |

## The TMD template (Friday's deliverable)

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# Technical Modeling Document – <feature>
**Sibling to:** PRD <link>, TCD <link>
**Authors:** <triad> | **Status:** Draft / Reviewed

## 1. Data model
Entity-level model: tables / collections, keys, relationships, indexes.
Storage choices and the trade-offs that drove them. (Day 1)

## 2. Cloud topology
Region / AZ choice, managed services, multi-tenancy stance. (Day 2)

## 3. API contract
Resource design, methods, versioning, idempotency, error semantics.
REST or SOAP – defended choice. (Day 3)

## 4. System sequence model
End-to-end happy-path sequence diagram + 1–2 sad/weird-path
sequences. Names every hop, protocol, and failure handler. (Day 4)

## 5. Performance baselines & monitoring
– Baselines: what we measure today (or expect at launch)
– Targets: from TCD Section 4 SLOs
– Monitoring plan: dashboards, alerts, on-call signal
– AI-summary validation log (this week's AI-assisted summaries
  with their validation status) (Day 5)
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## Triads

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Same triads from Weeks 1–4. PRD authors → TCD authors → TMD authors. By Week 6 these triads will negotiate this work with stakeholders.

## Friday review rubric (TMD)

| Dimension                   | Weight | What "exemplary" looks like                                                 |
|-----------------------------|--------|-----------------------------------------------------------------------------|
| Data model clarity          | 20%    | Entities, keys, indexes, and trade-offs all named; storage choice defended  |
| Cloud topology realism      | 15%    | Region/AZ stance fits the SLOs; multi-tenancy stance honest                 |
| API contract sharpness      | 20%    | Resources are nouns; idempotency, errors, versioning explicit               |
| Sequence model completeness | 20%    | Happy + sad/weird paths covered; protocols named; failure handlers shown    |
| Baselines + monitoring      | 15%    | Baselines real; monitoring fires before users notice                        |
| AI-validation discipline    | 10%    | Validation log distinguishes correct / partly correct / wrong with evidence |

## Bridge to Week 6

Week 6 (Stakeholder Alignment & Negotiation) takes the TCD's Stakeholder Sign-Off Matrix (Section 6) and the TMD's technical decisions and **negotiates** them. Each triad picks one or two constraints to actively negotiate with simulated stakeholders. The TMD's specificity is what makes those negotiations productive — vague briefs produce vague conversations.